

Part 1: Schema Creation & Constraints (DDL)

Create the table **CARGO_SHIPMENT** with the following specification:

Column Name	Data Type	Constraints / Description
Shipment_ID	VARCHAR2(10)	Primary Key
Flight_Code	VARCHAR2(10)	Cannot be null
Origin_Hub	VARCHAR2(15)	Must be uppercase only via check constraint
Dest_Hub	VARCHAR2(15)	Cannot be null
Category	VARCHAR2(20)	Restricted to: 'Pharma', 'Perishable', 'Electronics', 'Heavy Mach', 'General'
Weight_Kg	NUMBER(8,2)	Must be strictly greater than 0
Freight_Cost	NUMBER(10,2)	Must be non-negative
Booking_Date	DATE	Default is system date
Handling_Agent	VARCHAR2(25)	Nullable

SQL> create table CARGO_SHIPMENT(

```
2 Shipment_ID varchar(10) constraint shidcon primary key,
3 Flight_Code varchar(10) constraint flicodcon not null,
4 Origin_Hub VARCHAR2(15) constraint orgicon check(origin_hub = UPPER(origin_hub)),
5 Dest_Hub VARCHAR2(15) constraint deshubcon not null ,
6 Category varchar(20) constraint catcon check(category in('Pharma', 'Perishable',
'Electronics', 'Heavy Mach', 'General')),
7 weight_kg number(8,2) constraint weigcon check( weight_kg > 0),
8 Freight_Cost NUMBER(10,2) constraint freigcon check(Freight_Cost>0),
9 Booking_Date DATE default sysdate,
10 Handling_Agent VARCHAR2(25));
```

Table created.

INSERT VALUES

```
SQL> INSERT INTO CARGO_SHIPMENT VALUES ('SHP102', 'EK-504', 'DUBAI', 'KOCHI',
'Electronics', 1200.00, 185000.00, TO_DATE('2026-08-12', 'YYYY-MM-DD'), 'ApexGlobal');
```

1 row created.

```
SQL> INSERT INTO CARGO_SHIPMENT VALUES ('SHP103', 'AI-801', 'DELHI', 'KOCHI',
'Perishable', 950.00, 42000.00, TO_DATE('2026-08-12', 'YYYY-MM-DD'), 'SwiftAir');
```

1 row created.

```
SQL> INSERT INTO CARGO_SHIPMENT VALUES ('SHP104', 'QR-312', 'DOHA', 'MUMBAI',
'General', 150.00, 12500.00, TO_DATE('2026-08-15', 'YYYY-MM-DD'), NULL);
```

1 row created.

```
SQL> INSERT INTO CARGO_SHIPMENT VALUES ('SHP105', 'EK-504', 'DUBAI', 'KOCHI',
'Pharma', 320.00, 75000.00, TO_DATE('2026-08-18', 'YYYY-MM-DD'), 'ApexGlobal');
```

1 row created.

```
SQL> INSERT INTO CARGO_SHIPMENT VALUES ('SHP106', 'BA-119', 'LONDON', 'DELHI',
'Electronics', 640.00, 210000.00, TO_DATE('2026-08-20', 'YYYY-MM-DD'), 'SkyLog');
```

1 row created.

```
SQL> commit;
```

Commit complete.

```
SQL> set linesize 300;
SQL> select * from CARGO_SHIPMENT;
```

SHIPMENT_I	FLIGHT_COD	ORIGIN_HUB	DEST_HUB	CATEGORY	WEIGHT_KG	FREIGHT_COST	BOOKING_D	HANDLING_AGENT
SHP101	AI-801	DELHI	KOCHI	Pharma	450.5	68000	10-AUG-26	SwiftAir
SHP102	EK-504	DUBAI	KOCHI	Electronics	1200	185000	12-AUG-26	ApexGlobal
SHP103	AI-801	DELHI	KOCHI	Perishable	950	42000	12-AUG-26	SwiftAir
SHP104	QR-312	DOHA	MUMBAI	General	150	12500	15-AUG-26	
SHP105	EK-504	DUBAI	KOCHI	Pharma	320	75000	18-AUG-26	ApexGlobal
SHP106	BA-119	LONDON	DELHI	Electronics	640	210000	20-AUG-26	SkyLog

6 rows selected.

Alter table

Ddl :

Add an **INSURANCE_STATUS** column with a default value and a check constraint:

```
SQL> alter table CARGO_SHIPMENT add INSURANCE_STATUS varchar(20) default 'unpaid' constraint insucon check (INSURANCE_STATUS in ('paid','unpaid','waived'));
Table altered.
```

Check null value from handling agent

```
SQL> select * from cargo_shipment where handling_agent is null;
```

SHIPMENT_I	FLIGHT_COD	ORIGIN_HUB	DEST_HUB	CATEGORY	WEIGHT_KG	FREIGHT_COST	BOOKING_D	HANDLING_AGENT	INSURANCE_STATUS
SHP104	QR-312	DOHA	MUMBAI	General	150	12500	15-AUG-26		unpaid

Display each cargo category, count of shipments, and total weight

select category,count(SHIPMENT_ID), sum(WEIGHT_KG) from cargo_shipment group by category ;

```
SQL> select category,count(SHIPMENT_ID), sum( WEIGHT_KG) from cargo_shipment group by category ;
```

CATEGORY	COUNT(SHIPMENT_ID)	SUM(WEIGHT_KG)
Pharma	2	770.5
Electronics	2	1840
Perishable	1	950
General	1	150

Find the flight codes that carry a total weight of **at least 1000 Kg**:

```
SQL> select FLIGHT_CODE,sum(WEIGHT_KG) from cargo_shipment group by FLIGHT_CODE having sum(WEIGHT_KG) >1000;
```

FLIGHT_COD	SUM(WEIGHT_KG)
AI-801	1400.5
EK-504	1520